

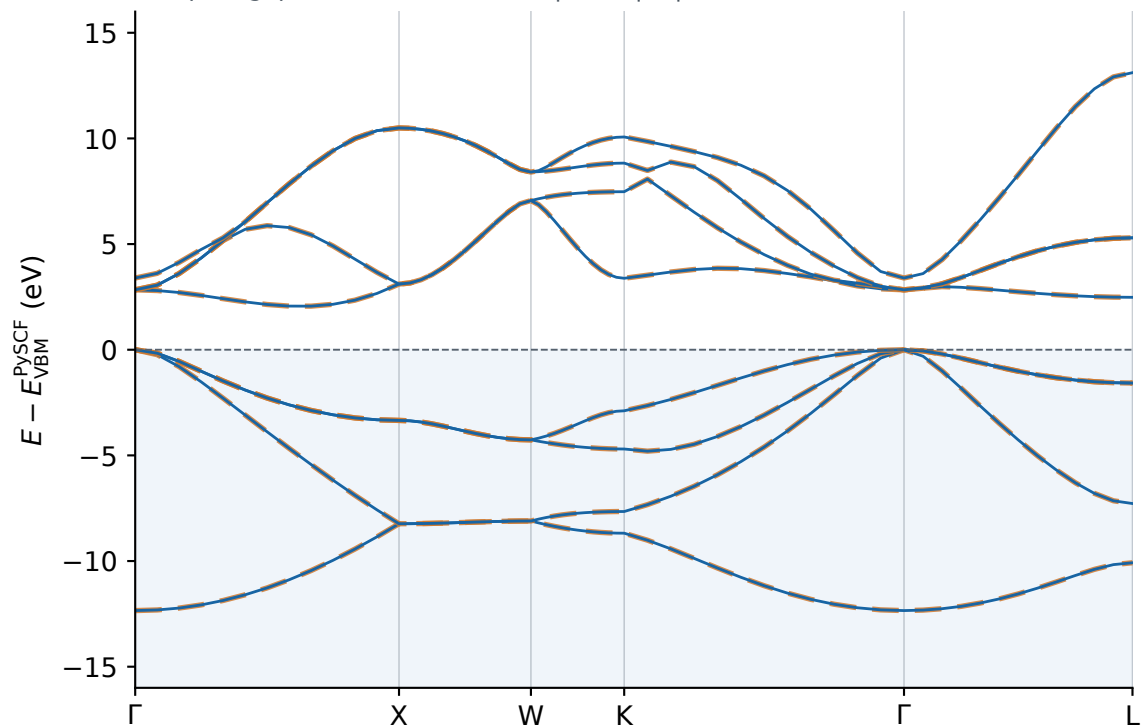
Periodic PBE bands: GradSCF vs PySCF

GTH-SZV / GTH-PBE • SCF mesh $2 \times 2 \times 2$ • FFT 41^3 • 61 path points, 8 bands

— GradSCF — PySCF

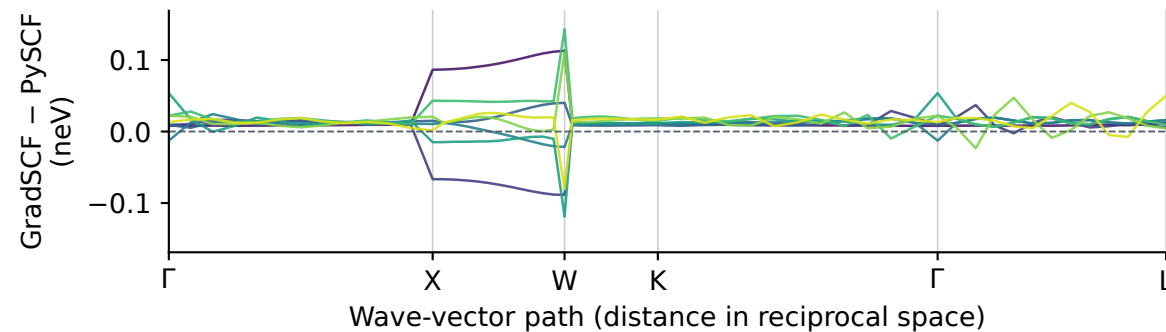
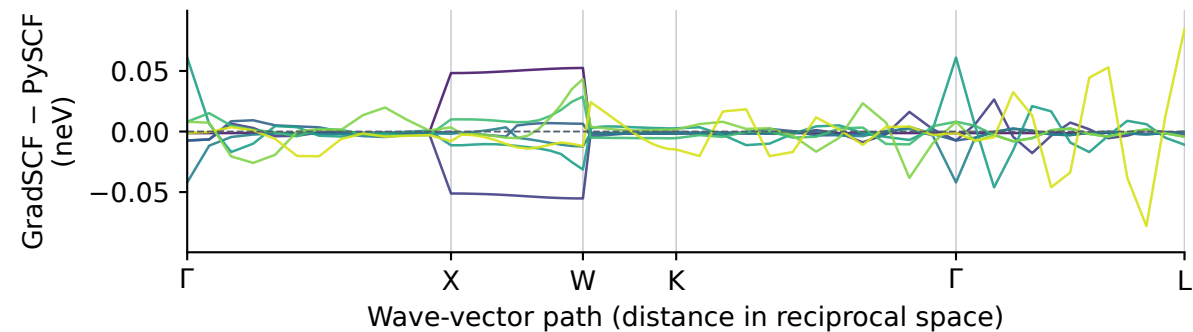
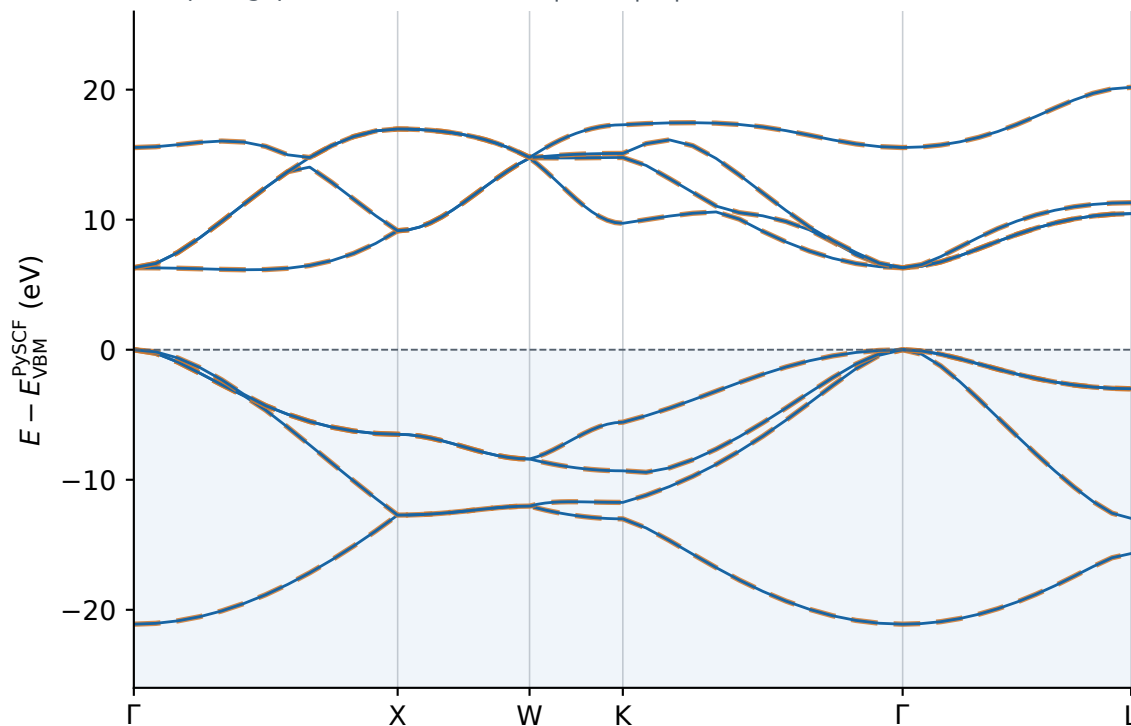
Silicon (Si) | $a = 5.430 \text{ \AA}$

Sampled gap: 2.0625 / 2.0625 eV | Max $|\Delta E| = 8.45\text{e-}11 \text{ eV}$



Diamond (C) | $a = 3.567 \text{ \AA}$

Sampled gap: 6.1503 / 6.1503 eV | Max $|\Delta E| = 1.43\text{e-}10 \text{ eV}$



One shared PySCF VBM zero per material. Sampled gap values: GradSCF / PySCF.

Minimal basis and coarse SCF k mesh; basis and k-mesh convergence of the sampled gaps remains untested.